Unit 9 - Computer Networks
Assignment 2 - Design a LAN Tasks 3&4

Introduction

Scenario / Vocational Context - *You are a junior consultant for T5 Alliance. A new client, Competition Bikes Ltd, are a new start-up company that manufacture a range of carbon fibre race and hybrid bikes from their small factory and sell them online. They have commissioned you to design and implement a new local area network with the following requirements:*

The network must support at least 100 client computers with room for 20% growth.
There are five departments: Management (8 staff), Accounting (4 staff), Engineering (70 staff), Sales & Marketing (10 staff) and IT (8).

The network topology should be designed to be fault tolerant and adhere to the "Core, Distribution, Access" model of network design.

5 LAN servers will connect to the master switch: 2 Domain Controllers, 1 File & Print Server, 1 Email server and 1 SQL Database Server

Network media should be provided as follows: Cat 6 UTP cable should connect the servers and PCs to the department switches. 1000BASE-SX fibre optic cables should connect the switches together.

There should be a DMZ that contains a further 3 servers: Web Server, FTP Server and a Remote Access Server.

The DMZ should provide the maximum security possible. Support for remote access must be included in your design.

Task 6

(U9D1) - Describe network technologies - Justify the design and choice of components used in a particular networked solution

Introduction

For this task, I will be explaining the justification of the different components that has been added to the network diagram which I made previously. This will include the switches, the connections and firewalls that I have recommended.

For the components added, I will be stating why I have chosen the components, also stating how they can help the business with specific tasks. The parts I will by justifying will be the number of switches used, the type of switches, the way the switches are connected, the number of firewalls used and the type of firewalls used.

Basically, I will be explaining the components I recommended and why I have recommended them, also explaining the advantages they can give to the company. This is to show that the component I have recommended can be useful for this type of network and for the company.

Justification of the number of switches used

In the network diagram, I have included 12 switches in total. Most of them (7 switches) acting as a connecting device to the clients within the network and an extra 2 acting as master switches, making distribution layers for the network. While the rest is used for different jobs, such as firewalls (2 switches) and one as a DMZ switch.

To elaborate, the purpose for the 7 switches used as a connecting device to the clients, are there to connect the PCs, used by the staff, to the network itself. The master switches are used as the distribution layer of the network, so the routing and filtering is managed by these switches.

In essence, the reason that I have chosen this many switches is due to the network architecture and how many clients are needed for the network. Which means having this many switches, makes the network work efficiently and it also has to do with the way the network has been architected in the diagram.

The advantages to having this many switches is that it has a lot of room for the estimated clients of the network and even for extra client growth. Basically, the more switches there are, the less clients there are connected to a single switch.

This is an advantage because it means each individual switch has less maintenance when compared to a switch with many clients connected to it. While this does add more switches to maintain, it theoretically reduces the maintenance time per switch, because of the smaller number of clients connected and that it would be very unlikely that two or more switches would malfunction at the same time.

Another advantage this can have is that having this many switches makes the network less prone to completely shutting down when one or two switches go down. In simple terms, the network is a lot less likely to stop the flow of information from one switch malfunctioning.

What I mean by this is that, by having a lot more switches to have clients connected to and connect to each other with, means that the information from other switches can still flow to a lot more clients, compared to a network with an equal amount of clients and less switches.

The disadvantages to having this many switches is that it can cause some time delay within the switching process, especially when the network is experiencing slow speeds. Meaning that the network will take a lot longer to respond to the client, reducing valuable time and work.

This brings onto another disadvantage, which is that having a lot of switches means checking whether they all remain in the same broadcast area. This can prove to be a disadvantage due to one or more switches failing to be in the same broadcast.

Which means that they will not be receiving information from other sources, which can hinder them significantly less useful. In order to control this, more maintenance time is needed on the network, which can again reduce time and work.

Overall, I believe that having this many switches in the network can prove to be very advantageous due to the reduction of a significant network malfunction, the efficient

maintenance of each switch and even the security that the very few switches can bring to the network.

Even though there are disadvantages to having a lot of switches, I believe that the advantages of having a more robust network outweigh the negatives of the network being somewhat slower and increased overall maintenance tasks for each switch.

Justification of the type of switches used

One of the reasons I have so many switches is due to the different jobs some of switches can have. While most of them have the jobs of connecting the client to the network, there are a few which have completely different jobs to the majority, and those jobs can prove to be significantly important to the network.

This means that most switches in the network have small differences, such as having more ports, or being completely different to each other, due to the jobs they carry out or due to the different manufacturers. Basically, the reason I have different types of switches is due to the many jobs that switches can carry out, due to the way they are designed and manufactured. Which means that there are some switches doing different operation to others.

As I have mentioned before, the purpose for the 7 switches used as a connecting device to the clients, the master switches are used as the distribution layer of the network, so the routing and filtering is managed by these switches and the final two firewall switches are used to control the connections to the network, but the firewalls will be mentioned afterwards.

In essence, the reason that these type of switches have been chosen is due to the importance that layer 2 switches and layer 3 switches can bring to the network. Which means that the network can be managed by an administrator, instead of having no manager of the network or everyone being a manager.

The advantages of having a layer 3 switch as a master switch and a layer 2 switch as the client switches is due to the data management the layer 3 switch offers and the ease of maintenance of the layer 2 switch offers. In simple terms, the network can be easily controlled from two distribution layers, instead of having many layer 3 switches with the ability to manage them.

What I mean by this is that, there really isn't a need to have more than two distribution layers, because I believe that having all the switches be a layer 3 switch wouldn’t be much of a benefit and really wouldn’t make much of a difference, apart from the fact it can make the data flow more complex, due to a layer 3 switch being a lot more specific than a layer 2.

So, only having 2 layer 3 switches controlling the flow of many layer 2 switches is a more efficient method due to data being easier to manage and so it doesn’t add more unnecessary complexities and extra maintenance to the network.

The disadvantages of having very little layer 3 switches control a lot of layer 2 switches is that, if the layer 3 switches cease to function, the network would be completely inaccessible to the clients, meaning that the layer 3 switches will need to be restored into working order or replaces.

However, the chances of this actually happening is very rare, especially when more than one layer 3 switch is installed in the network. So, having two layer 3 switches will half the chances of this actually happening.

Overall, I believe that having these types of switches within the network can prove to be very advantageous due to the importance that switches made to carry out a specific job, such as maintenance these switches can give.

Even though there are disadvantages which can impact the network significantly, I believe that the risks are extremely rare, and that there are some risks within a network which are impossible to completely remove, but they can be reduced to a point where they are so rare they might never actually happen.

Justification of the way in which your switches are connected to each other

In the network diagram, I have showcased the way that the switches will be connected to each other, by using both 1000BASE-SX fibre optic cables and Cat 6 UTP twisted pair cables, in a Star-Bus hybrid topology. With the fibre optic acting as the high speed connection between the switches and the distribution layers and the rest of the connections being the twisted pair cables.

Also, the reason that the network is laid out in this specific way is so the network can carry on flowing, even when a switch, server or even a master switch malfunctions. Basically, the network has been connected in a way where the data will carry on flowing, even if any of the switches or even servers has an error.

To elaborate, the purpose for the 1000BASE-SX fibre optic cables is so there is a high speed between the client switches and distribution layers, so there is a faster flow of information, and the Cat 6 UTP twisted pair cables are used to connect other devices to the network which doesn't need faster connection speeds.

In essence, the reason that I have chosen these type of cables and the layout of the network is due to the efficiency of the data flow of the network. Which means that the data and information can be delivered at decent speed and carry on being delivered even when one of the layer 2 or layer 3 switches cease to work within the network.

The advantage to having the network structured this way is, as I have mentioned before, is making the network less vulnerable to being completely or sometimes partially unusable when a component stops working. Basically, the connection each component has to each other has more than one way to send data.

This is an advantage because it means that the data from a switch is at a significantly lower risk to being completely cut off from the rest of the network. While this does add more cables to maintain, it technically reduces the chances of data not passing through the network, because of the larger number of connections, making it more unlikely to completely cut off the data flow of one switch.

The advantage of having fibre optic working between the distribution layer and the client switches and having the other cables as the slower but not too slow twisted pair cables, improves the speed in which the data is flowing and the speed in which the switch can be managed. In simple terms, the network can send data to the client switches a lot faster with

the fibre optic and the twisted pair cables are installed where fibre optic wouldn't make a difference.

What I mean by this is that, by having fibre optic installed where faster data flow is more important, means that the information from the distribution layer is significantly faster, compared to a network with a complete setup of twisted pair cables.

The disadvantages to having fibre optic installed in the network means that there are fragile and expensive cabling within the network, which means that they can break easily and cause many problems for the network and the company itself. Meaning that the costs of repairing or installing new fibre optic cables would not be cheap.

This brings onto another disadvantage, which is the fact that the fibre optic cables are on a vital part of the network, which means that the network will have to be temporarily unusable when the cables break until they are repaired. However, it is rare for this to happen, but since they are more fragile than a twisted pair, it is most likely for this to happen to fibre optic.

Which means that the company wouldn't be able to use the network for a specific amount of time, which can have significant impacts to the company. Even though I have connected the two distribution layers to every single switch, the cables are all still vulnerable to break, however the chances are completely lower than if I only connected one distribution layer to one switch. However, in order to check whether the fibre optic cables are working, more routine checks is needed on the cables, which can again reduce time and work.

Overall, I believe that having these types of cables within the network can prove to be advantageous due to the speed improvements that fibre optic cables bring to the network, making the flow of data and the speed of management significantly faster, due to the fact a lot of data is passed through the distribution layers and switches, so having twisted pair instead of fibre optic would slow down the network massively.

Also, the star-bus topology on the diagram helps with the connectivity of the switches, which has been mentioned before, but not directly to the topology. The way it helps is that each switch is connected to two different distribution layers thanks to this topology, meaning if one of the distribution layers or one or more of the layer 2 switches go down, the network can carry on working while they are being repaired.

Even though there are disadvantages which can impact the connection and cabling of the network significantly, I believe that the risks are rare. As I have mentioned before, there are some risks within a network which are impossible to completely remove, but they can be reduced to a point where they might never happen.

Justification of the number of firewalls used

As I mentioned before, in the network diagram, I have included 12 switches in total, but most of them aren't important since I have already explained their purpose. However, the rest which is used for different jobs, are the firewall switches, which consists of 2 switches and another one as a DMZ switch.

To specify, the purpose for the 2 firewall switches are used as a protective layer from the entire network to the WWW. The DMZ switch is used as the buffer between the LAN network and the rest of the internet.

Therefore, the reason that I have chosen these switches is due to the type of security that laid out on the network diagram. Which means having 2 firewalls and a DMZ switch, makes the network have very good security, especially when connected to a very unsecure network such as the WWW.

The advantage to having two firewall switches and a DMZ switch is so there is maximum security possible for a network this size. Basically, having two firewall switches with a DMZ switch in the middle makes it a lot harder for someone from the WWW accessing the entire network.

This is an advantage because it means that the data is at a significantly lower risk to having an outside interceptor removing or plagiarizing the data within the network. Especially when the network has both the firewall switches made by different manufacturers, so if they bypass a loophole in the first one, they have to go through the DMZ and the other firewall which wouldn't have that loophole.

Which bring us to another advantage which is that, not having this type of security can be catastrophic for a company if their valuable data is removed or stolen by an outside source, open for a competitor, because it would make the business lose an advantage of the competition, meaning that having these switches reduces the chances of this type of scenario happening.

The disadvantages to having this many firewall switches is that it can cause extra constant maintenance on the firewalls, so the firewalls don't have a malfunction without knowing. Meaning that the firewall maintenance checks would be one of the largest priorities of the network, causing more workload and leaving the less prioritising jobs alone, which could have an impact on the network since every bit of the network is somewhat important.

This brings onto another disadvantage, which is that if the firewall switches do have a malfunction or stop working due to the lack of power reaching them. This can prove to be a significant disadvantage of the entire network being at risk of being intercepted, even if one of the firewall switches stop working.

Which means that the company is vulnerable from attacks or interceptions from the unsecure network of the internet, which can cause the company to lose valuable data. In order to control this, more maintenance time is needed on the firewalls, which can again reduce time and work.

Overall, I believe that having two firewall switches and a DMZ in the network can prove to be very advantageous due to the reduction of a significant network interception, due to the the security that the switches can bring to the network.

Even though there are very significant disadvantages to potential disasters and even maintaining 2 firewall switches and a DMZ switch, I believe that it is also an advantage to have more firewall switches, because it offers a higher level of security, even though it gives a higher level of risk, but so can a single firewall switch or firewall software, so it is practically impossible to get rid of the disadvantages.

Justification of the type of firewall(s) used

On the network, the type of firewalls that are used are hardware firewalls instead of a software firewall. Also, both of the firewall switches are completely different to each other, this is due to the better security that is offered through this configuration. Basically, the network has two firewall switches and a DMZ switch, which are physical hardware instead of a virtual software, and all these switches have different manufacturers to increase security.

This means that all these switches have small differences, such as the software it uses, or how it operates completely, due to manufacturers making their products differently to carry out the same purpose as others.

In essence, the reason I have different types of switches as firewall switches is due to the extra security of having different software in each one can bring. Which means that going through one firewall, will be completely different when going through the next, especially with the DMZ in the way.

One advantage to this is that a hardware firewall is usually more sophisticated than firewall software and has more advantages. Meaning that this gives one of the highest possible security component in a network of this size, which is a very important factor of a small network.

The reason this is such an important advantage is that there really isn't a need to have more than two firewalls, because I believe that having more than two firewalls wouldn’t make much of a difference, apart from the fact it would add more unneeded security, cost and maintenance for a network this small.

Another advantage for installing these hardware firewall is that, as I have mentioned before, both of them aren’t from the same manufacturer. Meaning that they operate differently from each other, especially software wise.

This is an advantage due to the fact it is a lot harder to surpass two different firewalls. What I mean by this is that if a hacker can break through the first firewall using a weak spot, the same weak spot will be non existent on the second firewall if it is from another manufacturer, especially if there is a DMZ switch blocking the path.

So, having firewall switches made by different manufacturers has much more security when compared to having two firewall switches of the same manufacturer, which would make it a lot easier to get through them due to both having a similar weak spot.

The disadvantages to having firewall switches manufactured by different manufacturers is due to the fact that each manufacturer has small differences with how their switches are managed and operates, even though they carry out the same job. Meaning that more maintenance skill and time is needed to keep these switches in working order.

Which brings on to a massive disadvantage that, because of having two firewall switches made by different, if one of the firewall switches cease to function, the network would be a lot easier to access through the internet, meaning that the security would be completely reduced for a longer time due to it taking longer to repair different types of firewall switches.

However, again the chances of this actually happening is rare, especially when more than one firewall switch is installed in the network. So, having two firewall switches will half the chances of this actually happening, just like the distribution layers, but it is still possible.

Overall, I believe that having firewall switches within the network can prove to be very advantageous due to the importance of security to the network, to save the network from being abused from an outside source or the data being intercepted and stolen/corrupted.

Even though there are disadvantages which can impact the network's security, I believe that the risks are rare, and that there are some risks within a network which yet again are impossible to completely remove, but they can be reduced.

Conclusion

In conclusion, I believe that the way I have designed the network can prove to be very advantageous, efficient with the data flow within the network and that it fits well within the customer's' requirements. However, this does not mean the network is perfect, it just fits the requirements well.

With the number of switches used, I believe that I have used a good amount of switches due to the fact they all can connect together well, meaning that each layer 2 switch will be connected to two layer 3 master switches, making maintenance a lot easier.

With the type of switches used, I believe that the switches fits their job and purpose well, such as the layer 2 switches and layer 3 switches. Also, these switches have very good reputation, meaning that they will carry out the job in an efficient and sophisticated manner.

With the connection of the switches, I believe that the connections with the topology and the cables are nowhere near perfect, but it is good enough for the customer's requirements for the network, making sure that data isn't slowed down massively to a point of unusability.

WIth the number of firewalls used, like the other switches, I believe that the switches fits their job and purpose well, such as the DMZ switch and firewall switches. Also, like the previous switches, these switches have very good reputation, meaning that they will carry out the job in an efficient and sophisticated manner.

With the type of firewalls used, again like the previous switches, I believe that the firewall switches fits their job and purpose well, such as the DMZ switch and firewall switches. Again, these switches have very good reputation, meaning that they will carry out the job in an efficient and sophisticated manner.

Basically, I have researched on the type of network and network components to create this network diagram, making sure that it fits both the requirements set out for the customers by the company and the requirements for a basic network, such as security, data flow and speed.